

Neha Ram Singasan

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PROFESSIONAL SUMMARY

Security-focused software engineer with hands-on experience in application security, cloud security, and automation across government and enterprise environments. Proven ability to design secure architectures, automate threat detection and response, and build resilient cloud-native systems. Strong background in red team automation, secure software development, and DevSecOps practices.

EDUCATION

Bachelor of Science, Computer Science Goh Foundation Endowed Scholar, GIC Sparks & Smile Awardee, CGPA: 3.2/4.0 Singapore Management University	2023 - 2027 Singapore
Diploma of Education, Cybersecurity & Digital Forensics DSTA Digital Polytechnic Scholar, NYP Scholar, CGPA: 3.9/4.0 Nanyang Polytechnic	2020 - 2023 Singapore

WORK EXPERIENCE

Software Engineer Intern University Of Victoria, INSPIRE Program	Sept 2025 – Dec 2025 Victoria, BC, Canada
- Developed a secure, gamified educational web platform using React and Node.js for middle school students. - Designed RESTful APIs with secure authentication, role-based access control, and data protection best practices. - Integrated school district identity management systems to ensure compliant and secure access for students and staff.	
Application Developer Intern Accenture	May 2025 – Aug 2025 Singapore
- Supported the cloud migration of Singapore’s national tax system, ensuring data consistency across 1,500+ database tables. - Participated in secure SDLC processes including code reviews, QA testing, and CI/CD pipeline improvements. - Collaborated in Agile teams using Azure DevOps for sprint planning, version control, and automated deployments.	
Cybersecurity Engineer Defence Science and Technology Agency (DSTA)	Sept 2022 – Feb 2023 Singapore
- Designed and implemented a proof-of-concept to automate red team operations using Infrastructure as Code (Terraform), aligned with MITRE ATT&CK tactics. - Built a prototype tool that analyzed Terraform scripts to generate automated attack plans for simulated vulnerable AWS environments. - Demonstrated feasibility of automating reconnaissance and deployment, reducing manual setup time by 40%. - Collaborated with cybersecurity researchers to validate automation workflows and enhance adversary simulation frameworks.	

PROJECTS

IT Solution Architecture AWS, React, Flask, CI/CD Singapore Management University	2025
- Built a secure, cloud-hosted CRM platform for a banking client. - Integrated AWS Cognito for OAuth2 login and configured API Gateway + ALB for high availability. - Deployed automated CI/CD pipelines to improve deployment reliability and speed.	

9-Ball Pool Management Platform <i>Spring Boot, React, TypeScript, MySQL</i>	2024
Singapore Management University	
- Led a team of 4 to build a secure full-stack platform for pool ball match management. - Implemented OAuth2 authentication, two-factor authentication (2FA), CAPTCHA, and role-based access control. - Designed secure backend workflows and frontend integrations.	
CVE Mitigation <i>PowerShell, Splunk SOAR, Windows Defender</i>	2022
Nanyang Polytechnic	
- Analyzed and mitigated CVE-2022-21999 (Windows Print Spooler Escalation). - Built automated incident response workflows using Splunk SOAR and SIEM alerts. - Configured log-based detections and email alerting for rapid response.	
Application Security <i>Python, Flask</i>	2021
Nanyang Polytechnic	
- Developed a secure web application defending against OWASP Top 10 vulnerabilities. - Implemented secure input validation, OTP-based authentication, and SQL injection prevention mechanisms.	
Password Manager Web App <i>Python, Flask, Azure Cognitive Services</i>	2021
Nanyang Polytechnic	
- Developed a secure password manager with speech-to-text CAPTCHA and cloud backups - Integrated automated backup and secure authentication workflows.	

CERTIFICATIONS

AWS Certified Solutions Architect – Associate	2025
Google IT Automation with Python	2023
Certified Ethical Hacker (CEH)	2022
Computer Hacking Forensic Investigator (CHFI)	2022
AWS Cloud Practitioner	2021

ACHIEVEMENTS

Winning Team , Ensign InfoSecurity Hackathon (Ideathon), Singapore Defence Technology Summit	2023
2nd Place , GIT x PayPal Capture The Flag (CTF)	2023
Participant , Various School CTFs including STANDCON and Infosec CTF, Inter-Poly Hackathon: Lag and Crash	2022

TECHNICAL SKILLS

Languages | Python, JavaScript, Java, SQL, PHP

Frameworks | React, Django, Spring Boot, Flask, Node.js, Express.js

Developer Tools | Git, Docker, Visual Studio Code, PyCharm, Postman, JIRA

Libraries | Pandas, NumPy, Matplotlib, Bootstrap, TensorFlow

Cloud & DevOps: AWS, Azure, Docker, Azure DevOps, CI/CD Pipelines

Security & Automation: Splunk SOAR, SIEM (Splunk), OAuth2, TOTP, OWASP Top 10, Checkov, Burp Suite, Metasploit, Nmap, Wireshark, Selenium, VirusTotal